

ASMIT DATTA

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EDUCATION

New York University <i>M.S. in Computer Science</i>	New York, NY Expected 2028
Arizona State University <i>B.S. in Computer Science, Minor in Business</i>	Tempe, AZ May 2026

EXPERIENCE

Software Engineer Intern (Capstone) <i>WatchDNA</i>	Aug 2025 – Apr 2026 Ontario, Canada (Remote)
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- Set up the backend from scratch - databases, JWT authentication with role-based access, and the security middleware around it
- Built WatchDNA's store locator: an interactive Mapbox map with real-time filtering, an admin console for data operations, and a Python pipeline that discovers retailer endpoints on its own
- Shipped a companion app on the same infrastructure, reusing the auth, security, and data layers

Technical Officer <i>The AI Society at ASU</i>	Aug 2025 – Jan 2026 Tempe, AZ
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- Mentored four freshmen through their first ML project, an emotion detection system reading facial expressions and audio cues
- Wrote the materials for and ran workshops that made machine learning approachable to students with no technical background

Software Engineer Intern <i>MyYogaTeacher</i>	May 2025 – Jan 2026 San Diego, CA (Remote)
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- Shipped a voice-to-SQL pipeline on Whisper with streaming LLM responses that replaced the admin team's Ctrl+F-and-edit workflow
- Wrote the React Native client that streams biometric data in real time, with offline-first sync and time-series deduplication so nothing is lost when the network drops

Research Study Assistant <i>University of Delhi</i>	May 2024 – Aug 2024 New Delhi, India
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- Worked under Prof. R. P. Singh on optimizing Gaussian Process classifiers for high-dimensional data

Software Engineer Intern <i>Flickmatch</i>	May 2023 – Aug 2023 New Delhi, India
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- Revamped Flickmatch's website end to end, turning the new Figma designs into responsive React and Material-UI components across every page
- Wrote the state management as modular pieces rather than per-page, so the same logic carried across the site as it grew

PROJECTS

Pash lnkd.in/p/dCwX6kYm	2026 AWS S3/SQS, Demucs v4, CUDA
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- Split songs into vocals, drums, bass and instrumentals with Demucs v4 - uploads land in S3, jobs queue through SQS, and separation runs on a remote CUDA worker, 5-10x faster than on CPU

Lead Engine asmit.space/lead-engine	2026 Python, FastAPI, React, SQLAlchemy
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- Kept five scout agents running constantly across Zillow, Redfin, FSBO, expired listings and the Maricopa County Assessor, deduplicating by address and scoring every lead on motivation, ability and timeline with the reasoning written out - 2,129 leads sourced

Torii asmit.space/ky	2026 TypeScript, Bun, Hono, SQLite
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- Put an MCP proxy between AI agents and the paid tools they call, enforcing identity, policy and budget on every request: JWT agent identity, a JSON policy engine, and a SQLite ledger recording every call and every reason one was blocked
- Handled the x402 payment flow end to end - intercept the HTTP 402, check the wallet, pay, retry - with a budget tracker that reserves funds before the upstream call so two agents cannot overspend at once; the whole thing ships on two production dependencies

Portfolio asmit.space	Three.js, Blender, React
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- Modeled my own room in Blender and put it in the browser with Three.js - you walk up to the desk, click the machine, and every project opens as its own artifact instead of a list of links

SKILLS

Languages: Python, SQL, TypeScript, JavaScript, HTML/CSS, C++
Data & Cloud: AWS, Docker, CI/CD, Git, CUDA, PostgreSQL, Supabase, MongoDB, SQLite, Firebase, Pytest
Backend & APIs: REST, FastAPI, Node.js, Express.js, Flask, Bun, Hono, GraphQL, WebSocket, JWT, Role-based access control, SQLAlchemy
AI & Agents: MCP, x402, LangChain, OpenAI API, Claude API, Whisper, Pinecone, Autonomous agents, Tool use, Prompt engineering, spaCy
Frontend & Mobile: React, React Native, Expo, Tailwind, NativeWind, Material-UI, Three.js, Blender, Vite, Recharts, Electron, Figma, Mapbox